

```
#include <stdio.h>
#include <time.h>
```

```
int isSafe ( int board [][10], int row, int col, int n)
{
    int i, j;
    for (i=0; i<10; i++)
        if (board [row][i])
            return 0;
    for (i=row, j=col; i>=0 && j>=0; i--, j++)
        if (board [i][j])
            return 0;
    for (i=row, j=col; j>=0 && i<n; i++, j--)
        if (board [i][j])
            return 0;
}
```

```
int place (int board [][10], int col, int n)
{
    if (col >= n)
        return 1;

    for (int i=0; i<n; i++)
        if (isSafe (board, i, col, n))
            ;
}
```

```

        if (place(board, col+1, n))
            return 1;
        board[i][col] = 0;
    }

    return 0;
}

```

```

int printBoard (int board[10][10], int n)
{

```

```

    printf("\nBoard\n");
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
            printf(" %d ", board[i][j]);
        printf("\n");
    }
}

```

```

int main()
{

```

```

    int n;
    int board[10][10] = {0};
    printf("\nEnter the no. of queens  
to place: ");
    scanf("%d", &n);

```

```

    if (!place(board, 0, n))
    {

```

```

        printf("\nNo Solution Found");
        return 0;
    }
}

```

clock = start, end;

start = clock();

PrintBoard(board, n)

end = clock();

double time - taken = (double)

(end - start) /

CLOCKS_PER_SEC;

printf("In Execution time : %.f", time - taken);
return 0;

2

Output :

Enter the no. of queens : 5

Board

1	0	0	0	0
0	0	0	1	0
0	1	0	0	0
0	0	0	0	1
0	0	1	0	0

Execution Time : 0.00100